

Sketch-to-Photo Translation using Pix2Pix GAN and Zero-Shot Image Labeling with CLIP

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Abstract - Interpreting and converting sketches into meaningful visual content is often challenging, as the sketch data is minimal and abstract. To overcome this, I introduce a two-stage deep learning framework that improves the understanding of sketches. The first stage of the pipeline uses a Pix2Pix-based generative model that transforms sketches to realistic images. This stage reduces the gap between line sketches and real photos to a great extent. In the second stage of the pipeline, a pretrained vision language model is applied to assign semantic labels to the generated images. It does this by connecting visual features with descriptive text. This approach allows for accurate and consistent labeling without needing a lot of task-specific training data. By combining generative synthesis with strong image-to-text matching, our pipeline provides a scalable solution for sketch interpretation, while significantly cutting down on manual annotation tasks.

Index Terms - Sketch-to-Image Translation, Pix2Pix GAN, CLIP, Zero-Shot Classification, Vision-Language Models, Semantic Labeling, Deep Learning

I. INTRODUCTION

Sketch-based image understanding is a challenging task in computer vision because of the abstract and minimalistic nature of sketches. Unlike natural images, sketches lack important visual details like color, shading, and texture, which are essential for extracting features. Their reliance on line-based structure creates ambiguity, and the wide range of drawing styles adds to the inconsistency across samples. These difficulties lead to a significant gap between sketches and real-world images, making semantic interpretation and classification difficult. This study reveals a two-stage deep learning framework that combines generative modeling with multi-modal representation techniques to address these challenges. In the first stage, I used a Pix2Pix conditional Generative Adversarial Network (cGAN) to convert sketches into photorealistic images.

This process enhances the sparse input by adding believable visual features, such as texture, shading, and structural cues, which helps to reduce the gap between the sketch and image domains. In the second stage of the pipeline, instead of depending on the traditional supervised classifiers, I used a pre-trained vision-language model to assign semantic labels to the synthesized images. By embedding both visual outputs and textual class descriptions into a common feature space, the model enables direct matching between the input sketches and labels. This approach makes the annotation more flexible and scalable without the need for additional labelled data, making it particularly useful for datasets that are either unlabelled.

This union of GAN-enhanced sketch improvement and CLIP-enhanced semantic tagging enables a scalable, annotation-light pipeline for sketch understanding. Not only does it minimize the shortcomings of abstract sketch data, but it also circumvents task-specific supervised classifiers. This research highlights the strengths that generative models and pre-trained vision-language systems when it comes to filling modality gaps and allowing accurate, human-compatible image interpretation in low-detail environments such as sketches. OpenAI's CLIP model [1] was specifically chosen for its zero-shot capability, which allows flexible, label-free classification by aligning image and text embeddings without the need for initiating.

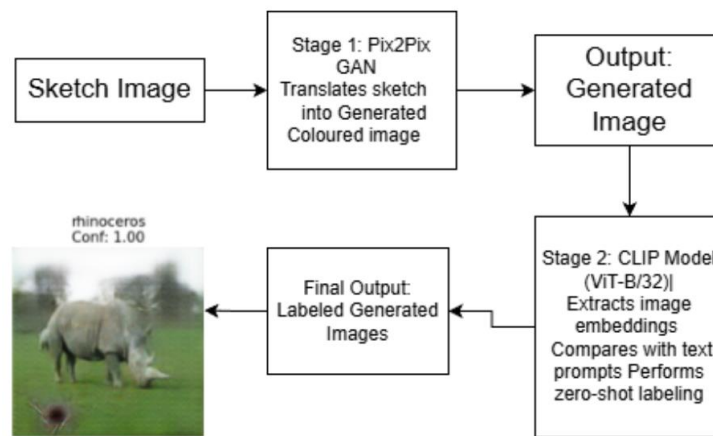


Fig.1 Two-Stage Image Labeling Pipeline

II. RELATED WORK

A. Sketch-Based Image Understanding

Sketch images are really difficult to interpret in the field of computer vision. This is due to the minimal and abstract nature of Sketches. Sketches typically consist of simple lines with usually some texture or shades, making it hard to extract information. Earlier methods usually struggle to adapt across different styles and domains. Recent efforts have often improved feature extraction and enabled more proper semantic understanding by narrowing down the difference between the sketch and the realistic image[1].

B. Generative Models for Domain Adaptation

Generative Adversarial Networks (GANs) have always showcased promising results in the domain of visual formats. Along with this, Pix2Pix [2] is an adopted supervised image-to-image translation framework that works over paired datasets. It employs a U-Net architecture for the generator to maintain consistency and a PatchGAN discriminator that focuses on local structural details. These features make the model effective for tasks like sketch-to-image translation, while retaining the original structure of the sketch. On the other hand, CycleGAN[3] supports unpaired data translation, offering more flexibility but needs more proper structural accuracy, which makes it less ideal to use in this situation where features need to be preserved

C. Vision-Language Models and Zero-Shot Learning

Combining visual and textual understanding has made a lot of progress in areas like zero-shot classification, image captioning, and content reclamation. The purpose of CLIP (Contrastive Language–Image Pretraining) [4] was to put images and their text descriptions together in a Joint embedding space. This means we can sort images without having to teach them how to do a specific job. The model can handle a lot of new inputs, like stylised or computer-generated images. This is useful when the data is either general or specific to a certain field. When used on outputs from generative models, CLIP's zero-shot learning is especially helpful because traditional supervised labelling may not be possible or may take a lot of time and money [5].

D. Hybrid Approaches: Generative Labeling Pipelines

To speed up image labelling and improve semantic understanding, more and more research has focused on combining generative models with large vision-language frameworks. This combined setup means making realistic images from simple or abstract sources, such as sketches or line drawings, and then using pretrained models like CLIP to give them labels. A pipeline like that would make labelling much more accurate and cut down on the need for manual annotation [6]. These hybrid methods, which are better at scaling and generalising than regular convolutional networks, have been especially useful in areas where there isn't much data or the inputs don't have very detailed visual features.

E. Feature Alignment and Preprocessing in Cross-Domain Learning

Cross-domain learning requires control of variation over the source and target domains to ensure effective extraction of features. Sketches and real images have vast structural difference and textures and require pathway steps like resizing, center cropping, and normalization. These mappings ease compatibility with models like CLIP, using a Vision Transformer (ViT) and being sensitive to input formats[7]. In addition to basic preprocessing, feature alignment techniques—such as perceptual loss from pre-trained models such as VGG—are employed to maintain semantic coherence during sketch-to-image translation. These retain better how generated images are anchored in downstream models. It has been shown that using such alignment techniques can greatly enhance the effectiveness of zero-shot classification tasks, particularly where data is sparse or artificially generated. [8].

III. DATASETS

The data set used in this work is a publicly available collection that is stored on Kaggle [9]. It is specially prepared for sketch-to-image translation tasks. It consists of paired examples of hand-drawn sketches and matching photorealistic images. The sketches are abstract and sparse, so the data set is suitable for experiments with generative models that recover detailed visual information from sparse input. The combination of sketches and photographs eases supervised training of conditional GANs such as Pix2Pix, increasing domain transfer and maintaining structure when synthesizing images. Every sample contains a paired sketch and photographic image to enable supervised learning methods. All images are stored in RGB color space to retain rich visual details and resize to 256×256 pixels. This consistent resolution guarantees compatibility across different deep learning architectures like convolutional neural networks of UNet and transformer-based models like Vision Transformers (ViTs)

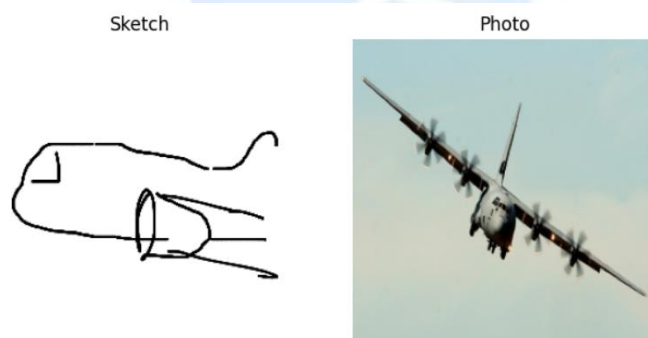


Fig. 2. Sketch Image and Real Photo for Aeroplane

The data is structured in six semantic classes with a single alphanumeric identifier. For each class, images are separated into two folders: sketch and photo. The structure supports simple pairing by class. One important aspect of the dataset is the imbalance in the sketch and photo classes. The sketch class is sparsely populated with around 75,481 sketches per class, providing a variety of hand-drawn inputs. Conversely, the photo class is restricted to just two of the six classes, each containing approximately 12,500 photo-realistic or GAN-generated images. The remaining four classes do not have photo counterparts at all, thereby introducing a stark asymmetry. This imbalance poses certain difficulties while training models, as models must learn to generalize well despite the absence of paired photo images in most classes.

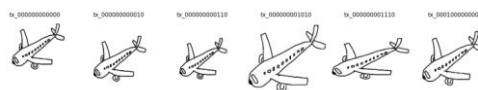


Fig. 3. Sample Sketch Images for 6 Classes

IV. PROPOSED METHOD

A. Pix2Pix GAN for Sketch-to-Image Translation

To translate sparse sketches into photorealistic images, I employ the Pix2Pix Generative Adversarial Network (GAN) [10], a supervised image-to-image translation framework. The objective is to minimize the domain gap between abstract sketch inputs and their corresponding photorealistic counterparts, facilitating downstream tasks such as classification and semantic understanding. Specifically, Pix2Pix is trained on aligned sketch-photo pairs using a U-Net-based generator and a PatchGAN discriminator, which promotes high-frequency detail reconstruction.

Training was conducted on an NVIDIA RTX A5000 GPU with 24 GB VRAM. The model was trained for 1000 epochs using 256×256 RGB image pairs. The version of the generator model that achieved the highest Peak Signal-to-Noise Ratio (PSNR) on the validation set was saved as the final model checkpoint for inference.

The generator follows an encoder-decoder architecture inspired by U-Net. Downsampling convolutional layers capture spatial hierarchies, while skip connections help retain essential features across scales. The bottleneck learns deep representations, and upsampling layers reconstruct full-resolution outputs. This design effectively preserves both global structure and fine details of the input sketch. Such architecture is particularly advantageous for sketch-to-image tasks, where maintaining edge integrity and spatial coherence is critical. Additionally, its symmetric design facilitates stable gradient flow, improving convergence during training.

The PatchGAN discriminator [10] evaluates overlapping image patches instead of the full image, enabling better enforcement of local realism and penalizing texture-level inconsistencies.

1) Loss Functions: The training objective combines multiple loss functions to balance structural fidelity, perceptual similarity, and realism: a) L1 Loss: Ensures pixel-wise accuracy between the generated image $G(x)$ and the ground truth y :

$$\mathcal{L}_{L1} = E_{x,y} [\|y - G(x)\|_1] \quad (1)$$

b) Adversarial Loss (GAN): Encourages the generator to produce images indistinguishable from real ones:

$$\mathcal{L}_{GAN} = E_{x,y} [\log D(x, y)] + E_x [\log(1 - D(x, G(x)))] \quad (2)$$

c) Perceptual Loss (VGG Feature Loss): Measures high-level feature similarity using a pre-trained VGG network ϕ :

$$\mathcal{L}_{perc} = \sum_l \|\phi_l(y) - \phi_l(G(x))\|_2^2 \quad (3)$$

where l refers to selected convolutional layers in the VGG network. The total loss used to train the Pix2Pix model is a weighted sum of the above components:

$$\mathcal{L}_{total} = \lambda_{GAN} \cdot \mathcal{L}_{GAN} + \lambda_{L1} \cdot \mathcal{L}_{L1} + \lambda_{perc} \cdot \mathcal{L}_{perc} \quad (4)$$

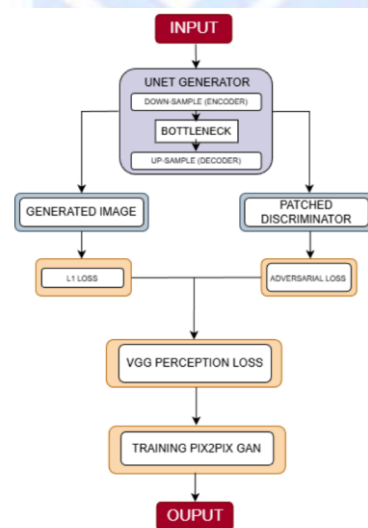


Fig. 4. Model Architecture

The generated images obtained from the trained Pix2Pixbased sketch-to-photo model are of resolution 256×256 pixels with three color channels (RGB), denoted as (3×256×256). This resolution strikes a practical balance between computational efficiency and visual clarity, making it well-suited for tasks like sketch-to-photo translation where fine edge details

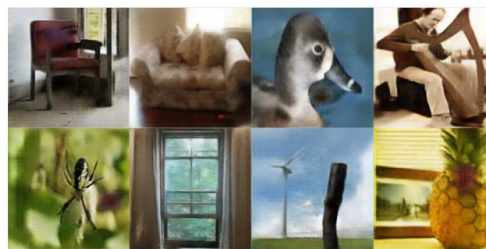


Fig. 5. Generated images after training

B. CLIP-based Zero-Shot Image Classification for Image Labeling

In this work, I utilized OpenAI's pretrained CLIP model, specifically the Vision Transformer-based ViT-B/32 architecture, which was trained on a large-scale dataset comprising 400 million image-text pairs to learn a unified embedding space for visual and linguistic modalities [12]. This model was selected for its strong zero-shot learning capabilities, allowing it to classify images without requiring any task-specific fine-tuning or labeled training data [13]. Rather than training a dedicated classifier, I framed potential labels as natural language prompts (e.g., "a sketch of a cat") and leveraged CLIP to compute cosine similarity between the image embeddings—derived from real or GAN-generated inputs—and the corresponding text embeddings. The label associated with the highest similarity score was selected as the predicted class. This strategy proved especially advantageous for automatically labeling images in domains with limited or no ground truth annotations, such as photorealistic reconstructions of sparse sketches [14].

During inference, each image is passed through the CLIP image encoder to obtain a normalized feature embedding. This embedding is then compared against all precomputed text embeddings using cosine similarity. The label corresponding to the highest similarity score is selected as the predicted class for that image. To provide a measure of certainty, similarity scores are converted into probabilities using a softmax function, and the associated confidence score is stored alongside the prediction.

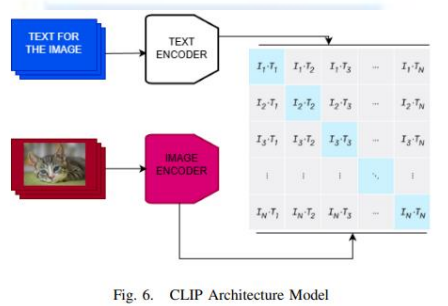


Fig. 6. CLIP Architecture Model

The future labels and their respective confidence scores are stored in a CSV file. This is convenient to examine, verify, or use later with assessment tools. To have a quick glance on how well the model performs in zero-shot labeling, we employ a straightforward visualization technique. It randomly select 10 labelled images and put them in a grid format for 2 cross 5. Each image in the grid depicts the label assigned to it. This is particularly helpful for seeing how well CLIP's comprehension works on AI-generated pictures.

To evaluate the effectiveness of the zero-shot labeling strategy, I generated a visual grid that displays a subset of GAN-generated images alongside their predicted labels. This visual analysis helps qualitatively assess how accurately the CLIP model associates visual patterns in the synthesized images with the intended semantic categories. In addition to visual inspection, I computed the Silhouette Score, a well-established clustering metric that quantifies how similar an image is to its assigned cluster compared to other clusters. In this context, a higher Silhouette Score suggests that CLIP's image embeddings form compact and well-separated clusters, indicating that the predicted labels align with semantically meaningful visual structures in the embedding space [15], [16].

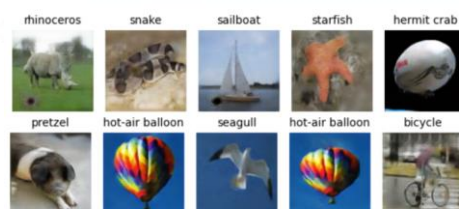


Fig. 7. Lablled Generated Images

This unsupervised measure is particularly valuable without ground-truth annotations, as it indicates a numerical measure of prediction reliability. Low or negative Silhouette Scores, on the other hand, are indicative of unclear content or semantic misalignment. To counteract these issues, I enhanced the text prompt creation by reformulating them to the sketch-based images' domain (e.g., "sketch of a cat") and normalized both text and image embeddings prior to calculating cosine similarity. This enhancement, coupled with using softmax scaling to translate confidence levels, reduced label noise and increased overall consistency of the predictions [17].

The t-SNE embedding of CLIP-created image embeddings provides a two-dimensional representation of the highdimensional semantic feature space learned by the model. The projection reflects the similarity with which the model clusters similar items. Well-segregated clusters indicate that the model detects semantic class differences, which result in uniform label assignment. In contrast, overlapping or diffused clusters could indicate label confusion, visual clutter, or semantic matching issues. When used together with clustering scores

such as the Silhouette Score, this visualization enhances our confidence in the alignment between the labels and the image content. It facilitates the evaluation of consistency and reliability of classification in instances where there are generated images or no prior labeled data.

V. RESULTS

A. Pix2Pix GAN to translate sketches into realistic images

I utilized the Pix2Pix GAN model to convert sketches into realistic images and verified its performance using PSNR and SSIM metrics. Throughout more than 1000 epochs of training, the PSNR consistently improved, reaching its peak at 22.97 at epoch 971. The model was improving in preserving pixel-level information. Similarly, the SSIM score continued to peak at 0.6290, suggesting that the model was also getting better at retaining the general structure and visual content of the images. The produced images are highly realistic. They retain key details from the original drawings and introduce credible textures and color. For instance, when a plain line drawing of a turtle was inputted, the model generated an image that had realistic water effects and features with clear lines that closely resembled a turtle. There are some small blurring issues, but the overall output indicates that the model can successfully transform simple sketches into realistic, photo-quality images. This exhibits its proficiency in sketch-to-image translation tasks.

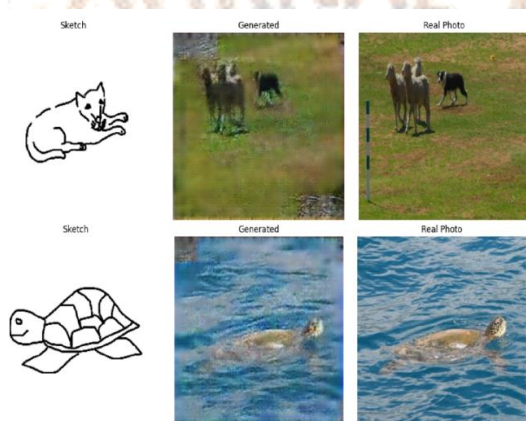


Fig. 8. Sketch vs Generated vs Real

B. CLIP-based Zero-Shot Image Classification for Image Labeling

The zero-shot classification outcomes for a set of generated images demonstrate the practical effectiveness of the CLIP model. Predictions included various object categories such as "squirrel," "couch," "pineapple," and "dolphin," with confidence scores ranging from 0.3779 to 0.9971. Visual inspection of these results confirms that CLIP correctly identified the content of high-quality generated images, such as a squirrel and a pineapple, both with full confidence. Lower confidence scores were observed for images with artifacts or reduced visual clarity, such as the dolphin prediction (0.38 confidence), likely due to ambiguity in image structure. Despite being trained without access to these specific generated samples, CLIP successfully associated visual content with appropriate textual labels, illustrating its strong generalization ability.



Fig. 9. Confidence score for labelled Images

To better understand the structure of the generated images, we applied t-distributed Stochastic Neighbor Embedding (t-SNE) to the image embeddings extracted using the CLIP model. The resulting two-dimensional projection, shown in Fig. 10, reveals moderately distinct clusters when colored by the predicted labels. This indicates that CLIP successfully captures underlying semantic similarities among images, even without task-specific fine-tuning. The formation of these clusters suggests that the model can effectively group visually or conceptually related images. To better understand the structure of the generated images, we used t-distributed Stochastic Neighbor Embedding (tSNE) on the image embeddings from the CLIP model. The resulting two-dimensional projection, seen in Fig. 10, shows moderately distinct clusters when colored according to the predicted labels. This clustering indicates that CLIP captures and encodes similarities between images, even without specific fine-tuning. To measure the quality of these clusters, we calculated the Silhouette Score. This score evaluates how well each data point fits within its cluster compared to others. It ranges from -1 to 1, where a higher value means the data points match their cluster well and fit poorly with nearby clusters. A Silhouette Score of 0.4105 shows a reasonable level of cohesion within clusters and clear separation between them.

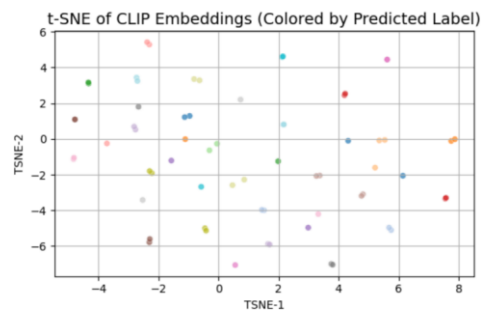


Fig. 10. T-SNE Of CLIP Embeddings

VI. CONCLUSION

This research proposes a framework that integrates generative modelling and pretrained vision-language models to solve the challenges with sketch-based comprehension. Sketches are typically abstract and texture-poor, making classification or semantic labelling challenging. To overcome this, the approach first employs a Pix2Pix-based Generative Adversarial Network (GAN) to produce realistic images from under-complete sketch inputs. These realistic images are then acted on by OpenAI's CLIP model, a large-scale, pretrained zero-shot classifier that imposes semantic labels based on visual-textual matching. There is no need for further tuning with CLIP, having been trained on over 400 million image-text pairs. I am using CLIP, which has been trained over 400 million image-text pairs, and removes the need for specific fine-tuning. This makes the system easily adaptable to new categories. Since CLIP is already familiar with a variety of image and text pairs, it manages to generate images well, even if they do not perfectly resemble real-world visuals. This method is useful in areas where labeled data is scarce or frequently changes, as it reduces the requirement for large, manually annotated datasets. In practice, this model can be quite useful across a wide range of applications. For example, it can label image datasets automatically, which reduces data preparation time. In sketch-based search systems, people would be able to sketch what they are looking for and get the corresponding results. It also has applications in content moderation, where coarse or abstract inputs must be quickly understood and classified. In education, it can assist tools that recognize and classify students' drawings. Professionally, illustrators and designers can turn hand sketches into labeled digital resources that enhance their workflow. The framework has some limitations. The generated images sometimes consist of visual relics that confuse the labelling process, and CLIP's accuracy relies on how prompts are supplied. Future enhancements could be achieved using more sophisticated image generation models like diffusion models or CLIP's optimization based on updated prompts. Future enhancements could be achieved using more enlightened image generation models like diffusion models or CLIP's optimization based on updated prompts. In conclusion, this complete pipeline offers a low-supervised and scalable solution for sketch-to-image conversion using a combination of a vision language tool and a pretrained model. It offers an exciting direction for building systems that reduce the manual labelling task while preserving less difference between abstract sketch and meaningful labels.

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